

Q1. Illuminated Area

Step 1: the swept height. Flip the sensor upside down so the illuminated band sits at the top. By the illumination lemma,

$$s(t^*) = 24 \text{ mm} \times \frac{\text{shutter speed}}{\text{flash sync time}} = 24 \text{ mm} \times \frac{1/200 \text{ s}}{14.73 \text{ ms}} = 24 \times \frac{5}{14.73} \approx 8.147 \text{ mm}.$$

Step 2: the paper's vertical extent. Both pieces are elevated 2 mm off the board, and the disk of radius 10 mm centred at $y = 12$ also has its lowest point at $y = 2$. Hence the paper occupies $y \in [2, 22]$, and its overlap with the illuminated band $[0, 8.147]$ is

$$y \in [2, 8.147].$$

Step 3: integrate. At each height y , the horizontal width of a strip is proportional to the square of its distance from the thin tip, so the three contributions are:

Strip 1 (broad head down). Width $\frac{(22 - y)^2}{50}$:

$$A_1 = \int_2^{8.147} \frac{(22 - y)^2}{50} dy = \frac{(20)^3 - (22 - 8.147)^3}{150} \approx 35.61 \text{ mm}^2.$$

Strip 2 (thin head down). Width $\frac{(y - 2)^2}{50}$:

$$A_2 = \int_2^{8.147} \frac{(y - 2)^2}{50} dy = \frac{(8.147 - 2)^3}{150} \approx 1.55 \text{ mm}^2.$$

Disk. Width $2\sqrt{100 - (y - 12)^2}$:

$$A_3 = \int_2^{8.147} 2\sqrt{100 - (y - 12)^2} dy \approx 81.96 \text{ mm}^2.$$

(The antiderivative is $(y - 12)\sqrt{100 - (y - 12)^2} + 100 \arcsin \frac{y - 12}{10}$; the numeric value follows from a CAS.)

Total illuminated area:

$$A = A_1 + A_2 + A_3 \approx 35.61 + 1.55 + 81.96 = 119.1 \text{ mm}^2.$$